

SUPPLY CHAIN & MANUFACTURING ANALYTICS PLATFORM

Project Type	End-to-end supply chain analytics consulting solution
Business Scope	Procurement, inventory, production, logistics and demand forecasting
Primary Technologies	SQL Server, Python, Power BI, SQL, statistical and machine-learning methods
Planning Horizon	Historical analysis: 2019-2024; predictive layer designed for forward periods

Prepared as a detailed implementation blueprint. Where no source data/model outputs were supplied, examples and targets are explicitly presented as proposed or illustrative rather than measured results.

1. Executive Summary

The proposed Supply Chain & Manufacturing Analytics Platform creates a centralized analytical layer for a mid-sized to large manufacturing organization experiencing rising logistics costs, inventory stock-outs, supplier delivery delays and production downtime. The solution connects procurement, inventory, production, logistics and demand planning into a common star-schema warehouse and exposes operational, predictive and executive views through Power BI.

The solution follows an end-to-end flow: raw CSV ingestion into staging, data quality and standardization, dimensional modeling, fact-table loading, analytical SQL views, statistical analysis, machine-learning models, model-output persistence in SQL Server, and business-facing dashboards. Each phase can be developed and validated independently before integration.

Expected business outcomes

- Improve visibility into supplier delivery performance, lead-time variability and supplier dependency.
- Reduce stock-out exposure by combining demand forecasts, current inventory, reorder levels and predicted stock-out probability.
- Identify logistics cost drivers by transport mode, shipment delay, order volume and route-related attributes when available.
- Quantify the operational relationship between delivery delays and production downtime.
- Create early-warning indicators for supplier delay, inventory imbalance and abnormal logistics cost or downtime events.
- Provide management with a common KPI language and drill-through capability from executive metrics to supplier, SKU, purchase order and shipment detail.

Important scope note: this report is a solution design and implementation blueprint based on the dataset description supplied. Actual KPI values, model accuracy and quantified savings must be calculated from the real CSV data; this report does not invent those results.

2. Business Problem, Objectives and Scope

Business problem

- Rising logistics costs are reducing margin and may be driven by transport mode, shipment volume, delays or inefficient routing.
- Frequent stock-outs indicate a mismatch between demand, inventory policy, replenishment lead time and supplier reliability.
- Supplier delivery delays create procurement uncertainty and can propagate into production scheduling and customer service.
- Production downtime needs to be linked to upstream supply events so management can prioritize high-impact operational risks.
- Existing data sources are fragmented, making it difficult to establish a consistent, auditable version of supply chain performance.

Project objectives

#	Objective	Primary deliverable
1	Build supply chain data warehouse in SQL Server	Staging, dimensions, facts and analytical views
2	Design ETL pipelines	Cleaning, standardization, validation and incremental load logic
3	Write complex SQL analytics	CTEs, windows, subqueries, ranking and KPI views
4	Develop Power BI dashboards	Executive, operations, supplier and SKU analysis

#	Objective	Primary deliverable
5	Perform statistical analysis	Hypothesis tests, regression, correlation and outlier analysis
6	Build predictive models	Demand, supplier delay and stock-out risk models
7	Integrate ML outputs into BI	Risk scores and forecasts persisted to SQL Server
8	Provide recommendations	Cost, inventory, supplier, logistics and production actions

3. Target Solution Architecture

The target platform uses a layered architecture that separates raw ingestion from curated analytics. This improves traceability, simplifies troubleshooting and allows business rules to evolve without rewriting source ingestion.

Layer	Purpose	Examples
Source	Operational flat files / future source systems	suppliers.csv, purchase_orders.csv, inventory.csv, production.csv, logistics.csv, sales_forecast.csv, calendar.csv
Staging	Raw landing with minimal transformation	stg_suppliers, stg_purchase_orders, stg_inventory, stg_production, stg_logistics, stg_sales_forecast, stg_calendar
Transform	Data quality, standardization and derived attributes	Date normalization, duplicate handling, delay calculation, validation flags
Warehouse	Conformed star schema	dim_date, dim_supplier, dim_sku, dim_warehouse, dim_transport_mode, fact_purchase_orders, fact_inventory_snapshot, fact_production, fact_logistics, fact_sales_forecast
Analytics	Reusable business logic	Supplier KPI views, inventory health, logistics trends, downtime impact, risk feature views
ML / Statistics	Model training and scoring	Forecasts, supplier delay probability, stock-out probability, statistical test outputs
BI	Decision support and early warning	Executive dashboard, operations dashboard, supplier/SKU drill-through

Logical flow

1. Ingest	2. Clean	3. Model	4. Analyze	5. Predict	6. Integrate	7. Act
CSV / source data	DQ + standardization	Star schema	SQL + Power BI	Python ML	SQL Server scoring tables	Alerts + decisions

Recommended operating principle: SQL Server remains the governed system of record for curated analytics and model outputs, while Python is used for statistical analysis, model development and scoring.

4. Data Model and Warehouse Design

The warehouse is designed as a star schema. Dimensions contain descriptive attributes, while facts contain measurable business events or periodic snapshots.

Table	Type	Grain	Key measures / attributes
dim_date	Dimension	One row per calendar date	Date, year, quarter, month, week, month start
dim_supplier	Dimension	One row per supplier	Supplier name, location, rating, segment
dim_sku	Dimension	One row per SKU	SKU and product attributes when available
dim_warehouse	Dimension	One row per warehouse	Warehouse identifier and location
dim_transport_mode	Dimension	One row per mode	Mode description
fact_purchase_orders	Fact	One row per PO line / order record	Order qty, promised date, actual date, lead time, delay days, on-time flag
fact_inventory_snapshot	Fact	One row per SKU-warehouse-date	Stock, reorder level, inventory status, stock-out flag
fact_production	Fact	One row per production record	Output quantity, downtime hours, downtime reason
fact_logistics	Fact	One row per shipment	Shipment cost, transport mode, transit delay, volume
fact_sales_forecast	Fact	One row per SKU-date forecast record	Historical demand, forecasted demand, forecast error

Recommended relationships

- fact_purchase_orders -> dim_supplier and dim_date; link to SKU if source contains SKU.
- fact_inventory_snapshot -> dim_sku, dim_warehouse, dim_date.
- fact_production -> dim_sku, dim_warehouse and dim_date where source keys permit.
- fact_logistics -> dim_supplier, dim_date and dim_transport_mode; retain shipment/order identifiers for traceability.
- fact_sales_forecast -> dim_sku and dim_date.

Data-modeling rule: preserve original business keys in each fact table even when surrogate keys are used. This supports reconciliation back to staging.

5. SQL Server DDL Blueprint

The following DDL is a representative implementation pattern. Column names should be adjusted to match the exact CSV headers after profiling.

```
CREATE TABLE dbo.dim_supplier ( supplier_key INT IDENTITY(1,1) PRIMARY KEY, supplier_id VARCHAR(50) NOT NULL UNIQUE,
supplier_name VARCHAR(200) NULL, location VARCHAR(200) NULL, supplier_rating DECIMAL(5,2) NULL, supplier_segment
VARCHAR(50) NULL, effective_start_date DATE NULL, effective_end_date DATE NULL, is_current BIT NOT NULL DEFAULT 1 );
CREATE TABLE dbo.dim_date ( date_key INT PRIMARY KEY, calendar_date DATE NOT NULL UNIQUE, year_num INT NOT NULL,
quarter_num INT NOT NULL, month_num INT NOT NULL, month_name VARCHAR(20) NOT NULL, month_start_date DATE NOT NULL,
week_num INT NULL ); CREATE TABLE dbo.fact_purchase_orders ( purchase_order_key BIGINT IDENTITY(1,1) PRIMARY KEY,
purchase_order_id VARCHAR(50) NOT NULL, supplier_key INT NOT NULL, order_date_key INT NOT NULL, promised_date_key INT
NULL, actual_delivery_date_key INT NULL, ordered_quantity DECIMAL(18,3) NULL, delivered_quantity DECIMAL(18,3) NULL,
delivery_delay_days INT NULL, on_time_flag BIT NULL );
```

6. ETL Pipeline Design

ETL objectives

- Load raw records without silently changing source meaning.
- Standardize dates, numeric fields, text keys and categorical values.
- Remove exact duplicates while preserving reconciliation counts.
- Flag invalid records rather than deleting them without traceability.
- Calculate delay days, on-time delivery, lead-time variability and stock-out status.
- Load dimensions before facts and quarantine records whose required keys cannot be resolved.

Step	Control	Example rule
1. File validation	Schema and file-level checks	Expected columns, encoding, file availability, row count
2. Staging load	Raw preservation	Load source values with ingestion timestamp and source file name
3. Standardization	Data types and categories	Trim text; convert dates; standardize transport-mode labels
4. Duplicate handling	Business-key uniqueness	Duplicate PO/shipment/SKU-date records reviewed
5. Data-quality rules	Validity	Actual delivery >= order date; cost >= 0; quantity >= 0
6. Derivations	Business metrics	Delay = actual delivery - promised delivery
7. Dimension load	Conformed keys	Upsert suppliers, SKUs, warehouses and modes
8. Fact load	Referential integrity	Resolve dimension keys and insert facts

Step	Control	Example rule
9. Reconciliation	Control totals	Source vs staging vs warehouse counts and sums
10. Audit	Traceability	Batch ID, load timestamp, rejected-record reason

```
SELECT TRIM(supplier_id) AS supplier_id, TRY_CONVERT(date, order_date) AS order_date, TRY_CONVERT(decimal(18,3),
ordered_quantity) AS ordered_quantity, TRY_CONVERT(date, promised_delivery_date) AS promised_delivery_date,
TRY_CONVERT(date, actual_delivery_date) AS actual_delivery_date, CASE WHEN actual_delivery_date IS NULL OR
promised_delivery_date IS NULL THEN NULL ELSE DATEDIFF(day, promised_delivery_date, actual_delivery_date) END AS
delivery_delay_days FROM dbo.stg_purchase_orders;
```

7. Data Quality and Validation Framework

Domain	Validation	Action
Dates	Order, promised and actual dates parse successfully	Quarantine invalid rows
Quantities	Ordered/delivered quantities are non-negative	Flag or reject by rule
Costs	Shipment cost is non-negative and numeric	Quarantine invalid values
Supplier keys	Every fact supplier resolves to dimension	Reject unresolved keys
Inventory	Stock and reorder levels are non-negative	Flag impossible records
Production	Downtime >= 0 and within plausible bounds	Outlier review rather than auto-delete
Forecast	Demand values follow business rules	Flag missing/negative demand
Duplicates	Business key uniqueness monitored	Deduplicate with documented rule

Recommended control dashboard

- Rows loaded / rejected / quarantined by file and batch.
- Duplicate count by source table.
- Missing-value rate by important field.
- Invalid date, cost and quantity counts.
- Referential-integrity failures.
- Source-to-warehouse row-count and monetary reconciliation.

8. SQL Analytics and Business Questions

The SQL layer should expose reusable views rather than embedding complex calculations independently in each Power BI visual.

8.1 Suppliers contributing to maximum delivery delays

```
WITH supplier_delay AS ( SELECT s.supplier_id, s.supplier_name, AVG(CAST(f.delivery_delay_days AS DECIMAL(10,2))) AS avg_delay_days, SUM(CASE WHEN f.delivery_delay_days > 0 THEN f.delivery_delay_days ELSE 0 END) AS total_delay_days FROM dbo.fact_purchase_orders f JOIN dbo.dim_supplier s ON f.supplier_key = s.supplier_key GROUP BY s.supplier_id, s.supplier_name ) SELECT * FROM supplier_delay ORDER BY total_delay_days DESC;
```

8.2 Top 3 delayed suppliers per month

```
WITH monthly_supplier AS ( SELECT d.year_num, d.month_num, s.supplier_id, s.supplier_name, AVG(CAST(f.delivery_delay_days AS DECIMAL(10,2))) AS avg_delay_days FROM dbo.fact_purchase_orders f JOIN dbo.dim_supplier s ON f.supplier_key = s.supplier_key JOIN dbo.dim_date d ON f.promised_date_key = d.date_key GROUP BY d.year_num, d.month_num, s.supplier_id, s.supplier_name ), ranked AS ( SELECT *, DENSE_RANK() OVER ( PARTITION BY year_num, month_num ORDER BY avg_delay_days DESC ) AS delay_rank FROM monthly_supplier ) SELECT * FROM ranked WHERE delay_rank <= 3 ORDER BY year_num, month_num, delay_rank;
```

8.3 Month-over-month logistics cost

```
WITH monthly AS ( SELECT d.month_start_date, SUM(f.shipment_cost) AS logistics_cost FROM dbo.fact_logistics f JOIN dbo.dim_date d ON f.shipment_date_key = d.date_key GROUP BY d.month_start_date ) SELECT month_start_date, logistics_cost, LAG(logistics_cost) OVER (ORDER BY month_start_date) AS prior_month_cost, logistics_cost - LAG(logistics_cost) OVER (ORDER BY month_start_date) AS mom_change, CASE WHEN LAG(logistics_cost) OVER (ORDER BY month_start_date) = 0 THEN NULL ELSE 100.0 * (logistics_cost - LAG(logistics_cost) OVER (ORDER BY month_start_date)) / LAG(logistics_cost) OVER (ORDER BY month_start_date) END AS mom_change_pct FROM monthly ORDER BY month_start_date;
```

8.4 Average shipment cost by transport mode

```
SELECT tm.transport_mode, AVG(f.shipment_cost) AS avg_shipment_cost, SUM(f.shipment_cost) AS total_shipment_cost, COUNT(*) AS shipment_count FROM dbo.fact_logistics f JOIN dbo.dim_transport_mode tm ON f.transport_mode_key = tm.transport_mode_key GROUP BY tm.transport_mode ORDER BY avg_shipment_cost DESC;
```

9. KPI Framework and Metric Definitions

KPI	Definition	Business interpretation
On-Time Delivery %	On-time purchase orders / eligible purchase orders * 100	Supplier delivery reliability
Average Delivery Delay	Average delay days using a documented treatment for early deliveries	Magnitude of delay
Inventory Turnover	COGS or consumption / average inventory	Inventory efficiency
Order Fill Rate	Delivered quantity / ordered quantity * 100	Supplier/service completeness
Logistics Cost %	Logistics cost / relevant sales, purchase or shipment value * 100	Logistics cost intensity
Production Downtime Ratio	Downtime hours / available production hours * 100	Operational availability
Lead-Time Variability	Standard deviation of observed lead time	Procurement uncertainty
Forecast MAPE	Mean absolute percentage error with zero-demand handling	Forecast accuracy
Stock-Out Rate	Stock-out SKU-date observations / eligible observations	Inventory service risk

Metric governance rule: every KPI must have a documented numerator, denominator, grain, date basis, missing-data treatment and refresh cadence.

10. Power BI Dashboard Design

Page	Purpose	Core visuals / interactions
Executive Overview	Management-level health of supply chain	KPI cards, cost trend, OTD trend, stock-out rate, downtime, risk summary
Supplier Performance	Identify delivery and dependency risk	Supplier ranking, average delay, OTD%, rating, monthly trend, drill-through

Page	Purpose	Core visuals / interactions
Inventory & SKU	Detect imbalance and stock-out exposure	Stock vs reorder, heatmap, SKU trend, forecast overlay, drill-through
Logistics	Control shipment cost and delay	Monthly cost, rolling 3-month cost, mode comparison, delay distribution
Production	Link upstream disruption to operations	Output, downtime ratio, downtime reason, delay vs downtime analysis
Risk & Forecast	Act on model outputs	Demand forecast, supplier risk, stock-out probability, threshold flags

Power BI measures

```
On-Time Delivery % = DIVIDE( CALCULATE(COUNTROWS(fact_purchase_orders), fact_purchase_orders[on_time_flag] = 1),
CALCULATE(COUNTROWS(fact_purchase_orders), NOT ISBLANK(fact_purchase_orders[on_time_flag])) )
Rolling 3M Shipment Cost =
CALCULATE( [Total Shipment Cost], DATESINPERIOD( dim_date[calendar_date], MAX(dim_date[calendar_date]), -3, MONTH ) )
```

For rolling metrics, ensure the date table is marked as a date table and that the relationship to the fact is active and correct. If a visual should show separate series, verify legend/axis configuration and whether the measures return different values in the current filter context.

11. Supplier Dependency, Inventory Imbalance and Composite KPI Ranking

Supplier dependency should not be defined solely by purchase value. A more useful view combines spend or order volume with SKU concentration, lead-time variability and supplier alternatives.

Measure	Suggested calculation	Interpretation
SKU dependency	Count of active SKUs sourced from supplier / total relevant SKUs	Breadth of dependency
Single-source SKU count	SKUs with exactly one active supplier	Continuity risk
Supplier concentration	Supplier share of purchase quantity/value	Commercial dependency
Lead-time volatility	Standard deviation of lead time	Planning uncertainty
Delay exposure	Share of orders delayed or total delay days	Execution risk

Composite supplier KPI

A transparent composite score can combine OTD, order fill, rating, delay and lead-time variability after stakeholder agreement on weights. Weights should be documented, tested for sensitivity, and presented alongside component metrics so users can understand score changes.

This report intentionally does not assign supplier scores because the actual dataset and agreed business weights were not provided.

12. Statistical Analysis Plan

Question	Method	Inputs	Output
Do supplier delays affect production downtime?	Two-sample test / regression with controls	Delay indicators, downtime hours, operational controls	Effect estimate, CI, p-value
What drives logistics cost?	Multiple linear regression	Cost, transport mode, delay, volume	Coefficients, R2, residual diagnostics
Does lead time relate to stock-outs?	Pearson/Spearman + regression	Lead time, stock-out frequency	Correlation and significance
Which logistics costs are abnormal?	IQR / z-score / robust method	Shipment cost by comparable segment	Outlier flag
Which downtime events are abnormal?	IQR / robust threshold	Downtime hours by context	Extreme incident list

Hypothesis-testing example

H0: supplier delivery delay is not associated with production downtime after accounting for relevant operational factors.

H1: supplier delivery delay is associated with production downtime. Report test/model, sample size, effect size, confidence interval and p-value. Statistical significance should not be treated as proof of causality.

Regression specification example

```
logistics_cost = beta0 + beta1 * shipment_delay_days + beta2 * order_volume + beta3 * transport_mode + beta4 * month/season controls + error
```

For categorical transport modes, use one-hot encoding with a documented reference category. Check multicollinearity, residual behavior and heteroscedasticity before interpreting coefficients.

13. Predictive Analytics Strategy

Use case	Target	Candidate models	Primary evaluation
Demand forecasting	Future SKU demand	Time-series baseline, regression, Random Forest where appropriate	RMSE, MAE, MAPE/WAPE, forecast bias
Supplier delay prediction	Probability an order will be late	Logistic Regression, Random Forest	Accuracy, precision, recall, F1, ROC-AUC; business impact
Stock-out risk	Probability of stock-out in horizon	Logistic Regression, Random Forest	Precision/recall, ROC-AUC, calibration, avoided stock-outs

Feature engineering

- Supplier: historical OTD%, average delay, delay variance, rating, recent trend.
- Purchase order: order quantity, order month, promised lead time, historical supplier behavior.
- Inventory: stock on hand, reorder level, days of supply, recent stock trend.
- Demand: lagged demand, rolling averages, seasonality, forecast error history.
- Logistics: transport mode, recent delay, shipment volume, historical cost, month/season.
- Production: recent downtime and output where the prediction point precedes the outcome.

Avoid target leakage: features must use only information available at the time the prediction would actually be made.

Example Logistic Regression workflow

```
from sklearn.pipeline import Pipeline from sklearn.preprocessing import OneHotEncoder, StandardScaler from sklearn.compose import ColumnTransformer from sklearn.linear_model import LogisticRegression from sklearn.metrics import roc_auc_score preprocess = ColumnTransformer([ ("num", StandardScaler(), numeric_cols), ("cat",
```

```
OneHotEncoder(handle_unknown="ignore"), categorical_cols) ]) model = Pipeline([ ("prep", preprocess), ("clf",
LogisticRegression(max_iter=1000, class_weight="balanced")) ]) model.fit(X_train, y_train) prob =
model.predict_proba(X_test)[: , 1] auc = roc_auc_score(y_test, prob)
```

14. Demand Forecasting Methodology

A practical forecasting design should begin with a baseline before introducing machine learning. The baseline may be last-period demand, moving average or seasonal naive. The model must demonstrate improvement over the baseline, not simply produce lower training error.

Phase	Action
1	Profile demand by SKU and date; identify missing dates, zeros, spikes and intermittent demand.
2	Create a complete date-SKU panel where business rules require continuous periods.
3	Create lag and rolling features using only historical observations.
4	Use time-based train/validation/test splits; do not randomly shuffle future periods into training.
5	Evaluate baseline and candidate models with RMSE plus MAPE/WAPE where appropriate.
6	Translate forecast performance into stock-outs, excess inventory and service impact.
7	Persist future forecasts and model metadata into SQL Server for Power BI consumption.

15. Integration of Predictive Outputs Back into SQL Server and Power BI

Recommended scoring tables

Table	Grain	Fields
ml_demand_forecast	SKU-date	sku_key, forecast_date_key, forecast_value, lower_bound, upper_bound, model_version, scored_at
ml_supplier_delay_risk	Supplier-order	supplier_key, purchase_order_id, risk_probability, risk_band, model_version, scored_at
ml_stockout_risk	SKU-warehouse-date/horizon	sku_key, warehouse_key, score_date_key, stockout_probability, risk_band, model_version, scored_at

Integration flow

Python training -> model artifact / scoring job -> SQL Server ML output tables -> Power BI semantic model -> dashboard thresholds / drill-through -> operational action -> outcome tracking

Risk bands should be based on validated thresholds. Thresholds can be selected using precision/recall trade-offs and business costs of false positives versus false negatives; exact values require real data.

16. Business Recommendations Framework

Area	Data-driven action	Expected business mechanism	Measurement
Supplier management	Segment suppliers by delivery reliability, dependency and criticality; target corrective action where risk is concentrated.	Lower disruption and improve service consistency.	OTD%, delay days, lead-time variability, expedite orders
Supplier rationalization	Identify single-source and highly concentrated SKUs; qualify alternatives for critical items.	Reduce continuity risk without blindly increasing supplier count.	Single-source SKU count, concentration, service level
Inventory optimization	Recalculate reorder points and safety stock using demand and supplier lead-time variability.	Reduce stock-outs while controlling excess inventory.	Stock-out rate, days of supply, inventory value
Logistics optimization	Compare cost and delay by mode and lane where available; investigate high-cost outliers.	Reduce avoidable freight and expedite cost.	Cost/shipment, cost/unit, delay rate
Production planning	Use supplier delay risk and demand forecasts in production scheduling and material planning.	Reduce downtime caused by upstream shortages.	Downtime ratio, shortage incidents, schedule adherence
Early warning	Surface supplier, inventory and logistics risk scores with owner and action.	Move from retrospective reporting to proactive intervention.	Alert precision, response time, avoided incidents

Quantifying business impact

- Savings from lower logistics cost = baseline logistics cost - post-intervention cost, adjusted for volume/mix where needed.
- Inventory benefit = reduction in stock-out losses plus reduction in excess inventory carrying cost, subject to finance validation.
- Supplier benefit = avoided expedite costs, avoided production disruption and improved service against a baseline.
- Forecast benefit = change in forecast error and resulting change in inventory/service outcomes.
- Every benefit claim should identify baseline period, comparison period, volume/mix controls and financial owner.

17. Implementation Roadmap

Sprint / Phase	Key activities	Exit criteria
1. Data profiling	Inspect CSVs, keys, date ranges, missingness and duplicates	Data dictionary + profiling report
2. Warehouse foundation	Create staging, dimensions, facts, indexes	DDL deployed and mapping approved
3. ETL	Build transformation, validation and reconciliation	Repeatable load with audit counts
4. SQL analytics	Create KPI views and complex business queries	Queries reconcile to controlled samples
5. Power BI	Build semantic model, measures, dashboards and drill-through	Dashboard UAT completed
6. Statistical analysis	Hypothesis tests, regression, correlation, outliers	Notebook with reproducible results
7. Predictive ML	Train, validate, tune and compare models	Model evaluation and scoring pipeline
8. Integration	Persist forecasts and risk scores; refresh BI	End-to-end scoring-to-dashboard flow
9. Recommendations	Translate findings into prioritized actions and impact model	Executive action plan
10. Presentation	Prepare final narrative and demo	Final presentation and handover

18. Testing, Governance and Operationalization

Test category	Examples
Unit testing	Transformation formulas, delay calculation, flags, KPI measures
Data reconciliation	Row counts, quantities, costs and key totals from source -> staging -> warehouse
Referential integrity	No unresolved dimension keys in production facts

Test category	Examples
Dashboard testing	Filter propagation, totals, drill-through and date behavior
Model testing	Time-based split, leakage checks, baseline comparison, calibration and threshold analysis
Refresh testing	Scheduled refresh, failed-load handling, incremental load behavior
Security	Role-based access if supplier or commercial information requires restricted views

Governance recommendations

- Maintain a data dictionary with source field, target field, data type, definition and transformation rule.
- Version SQL scripts, notebooks, Power BI files and model versions.
- Record model training date, feature set, evaluation window and threshold configuration.
- Use batch IDs and audit timestamps for every ETL run.
- Document data-quality exceptions and ownership.
- Review model drift and KPI definition changes periodically.

19. Final Deliverables Checklist

Deliverable	Contents	Status for this report
ER / star schema diagram	Dimensions, facts and relationships	Design specified; physical diagram after source profiling
SQL DDL scripts	Staging, dimensions, facts, indexes	Blueprint included; finalize against exact headers
ETL scripts	Cleaning, transformations, validation, loads	Design and representative SQL included
SQL analytics file	Complex queries and analytical views	Representative business queries included
Power BI PBIX	Executive and operational dashboards	Dashboard architecture and measures specified
Statistical notebook	Tests, regression, correlation, outliers	Methodology specified
Predictive notebooks	Forecast, supplier delay, stock-out models	Modeling framework specified
ML output tables	Forecasts and risk scores in SQL Server	Schema specified
Business recommendations	Prioritized actions + quantified impact	Framework specified; values require actual data
Final presentation	Executive story and demo	Suggested structure below
Executive summary	One-page management brief	Report contains executive summary; final numerical results require execution

20. Suggested Final Presentation Structure

Slide	Message
1. Business challenge	Rising logistics cost, stock-outs, supplier delays and downtime create connected operational risk.
2. Solution architecture	Centralized warehouse + ETL + analytics + ML + Power BI.
3. Data model	Show star schema and key facts/dimensions.
4. Supplier performance	Show delay and OTD analysis; identify concentration and critical suppliers.
5. Inventory health	Show stock-out exposure, imbalance and forecast-driven replenishment.
6. Logistics cost	Show monthly trend, rolling 3-month cost and transport-mode analysis.

Slide	Message
7. Production impact	Show relationship between upstream delays and downtime.
8. Statistical findings	Present effect estimates, significance and limitations.
9. Predictive models	Compare baseline vs selected models and business metrics.
10. Early warning	Demonstrate supplier delay and stock-out risk integrated into BI.
11. Recommendations	Prioritize supplier, inventory, logistics and production actions.
12. Business impact	Quantify validated savings/service improvements from actual data.

Appendix A. Proposed Data Dictionary

Source	Field	Type	Business meaning
suppliers.csv	supplier_id	VARCHAR	Unique supplier business identifier
suppliers.csv	supplier_name	VARCHAR	Supplier name
suppliers.csv	location	VARCHAR	Supplier location
suppliers.csv	rating	DECIMAL	Supplier rating
purchase_orders.csv	purchase_order_id	VARCHAR	Purchase order identifier
purchase_orders.csv	order_date	DATE	PO creation date
purchase_orders.csv	ordered_quantity	DECIMAL	Quantity ordered
purchase_orders.csv	promised_delivery_date	DATE	Supplier promised date
purchase_orders.csv	actual_delivery_date	DATE	Observed delivery date
inventory.csv	sku	VARCHAR	Stock keeping unit
inventory.csv	warehouse	VARCHAR	Warehouse identifier
inventory.csv	stock	DECIMAL	Inventory quantity at snapshot
inventory.csv	reorder_level	DECIMAL	Configured replenishment trigger
production.csv	production_id	VARCHAR	Production record identifier
production.csv	output_quantity	DECIMAL	Production output
production.csv	downtime_hours	DECIMAL	Downtime hours
production.csv	downtime_reason	VARCHAR	Downtime category
logistics.csv	shipment_id	VARCHAR	Shipment identifier
logistics.csv	shipment_cost	DECIMAL	Shipment/freight cost
logistics.csv	transport_mode	VARCHAR	Transport mode
logistics.csv	transit_delay	DECIMAL	Observed transit delay
sales_forecast.csv	sku	VARCHAR	SKU
sales_forecast.csv	historical_demand	DECIMAL	Observed historical demand
sales_forecast.csv	forecasted_demand	DECIMAL	Source forecast
calendar.csv	date	DATE	Calendar date

Appendix B. Additional Analytical SQL Patterns

Supplier performance ranking

```
WITH kpi AS ( SELECT s.supplier_id, s.supplier_name, 100.0 * AVG(CAST(f.on_time_flag AS INT)) AS on_time_pct,
AVG(CAST(f.delivery_delay_days AS DECIMAL(10,2))) AS avg_delay,
STDEV(CAST(DATEDIFF(day,d1.calendar_date,d2.calendar_date) AS DECIMAL(10,2))) AS lead_time_std FROM
dbo.fact_purchase_orders f JOIN dbo.dim_supplier s ON f.supplier_key=s.supplier_key JOIN dbo.dim_date d1 ON
f.order_date_key=d1.date_key LEFT JOIN dbo.dim_date d2 ON f.actual_delivery_date_key=d2.date_key GROUP BY
s.supplier_id,s.supplier_name ) SELECT *, RANK() OVER (ORDER BY on_time_pct DESC, avg_delay ASC) AS supplier_rank FROM
kpi ORDER BY supplier_rank;
```

Inventory imbalance flag

```
SELECT i.sku_key, i.warehouse_key, d.calendar_date, i.stock_quantity, i.reorder_level, CASE WHEN i.stock_quantity <= 0
THEN 'Stock-out' WHEN i.stock_quantity < i.reorder_level THEN 'Below Reorder' WHEN i.stock_quantity > 3 *
NULLIF(i.reorder_level,0) THEN 'Potential Excess' ELSE 'Healthy' END AS inventory_status FROM dbo.fact_inventory_snapshot
i JOIN dbo.dim_date d ON i.date_key=d.date_key;
```

The 3x reorder-level threshold is illustrative only. Production thresholds should use demand, lead time, service-level policy and carrying cost.

Appendix C. Model Evaluation Template

Model	Train window	Validation window	Test window	Metric 1	Metric 2	Business metric
Forecast baseline	Historical period	Later period	Holdout period	RMSE	MAPE/WAPE	Stock-out / excess inventory
Regression forecast	Historical period	Later period	Holdout period	RMSE	MAPE/WAPE	Bias / service impact
Logistic Regression	Historical orders	Later orders	Holdout orders	ROC-AUC	F1 / recall	Avoided late orders at threshold
Random Forest	Historical orders	Later orders	Holdout orders	ROC-AUC	F1 / recall	Avoided late orders at threshold

Model selection should consider business cost, interpretability, stability and operational usability rather than a single statistical metric. Thresholds should be tuned on validation data and frozen before final test evaluation.

Appendix D. Evaluation Rubric Mapping

Evaluation area	Weight	How this project addresses it
Data Modelling & ETL	20%	Layered staging-to-star-schema design, data quality rules, surrogate keys, reconciliation and ETL blueprint.
SQL Complexity	15%	CTEs, window functions, ranking, LAG, joins, KPI views and business questions.
Dashboard & Storytelling	20%	Executive, supplier, inventory, logistics, production and risk pages with drill-through.
Predictive Modelling	25%	Demand forecasting, supplier delay classification and stock-out risk with time-aware validation.
Business Insights	15%	Supplier, inventory, logistics and production recommendations linked to measurable KPIs.
Presentation	5%	Suggested 12-slide executive narrative and demo flow.

21. Conclusion

The proposed platform converts fragmented supply chain files into a governed analytical product that can support operational decisions and executive management. Its core strength is the connection between descriptive analytics and predictive decision support: supplier performance is linked to lead-time variability, inventory exposure, logistics cost and production disruption rather than being analyzed as isolated functions.

The immediate next step is to profile the seven source files and confirm exact columns, keys, date coverage, grain and data-quality characteristics. Once that profiling is complete, the DDL, ETL mappings, KPI formulas and model features can be finalized against real data. The final project should then demonstrate measurable changes in forecast accuracy, stock-out risk, supplier service, logistics cost and production downtime using actual model outputs and controlled comparisons.

End of report